

Section 2.4: Output Devices

Once data has been input into a computer system and processed, it is now 'useful information'. This information is ready to be output (sent out).

In this section you will learn about the different types of output devices that are commonly in use.

For each out device you have to be able to **identify it from an image**, explain what it is **used for** and discuss **advantages/disadvantages**.

Key objectives of this section:

- # Know the **purpose of** and be able to **identify** different output devices.
- # Understand the **uses of** each output device.
- # Be able to **discuss advantages and disadvantages** of each output device.

Output Devices (what are they?)

Definition: "Hardware devices that allow information to be sent out of a computer system."



Key Words:

Input, Processor, Output, Storage, Hardware, Data, Information

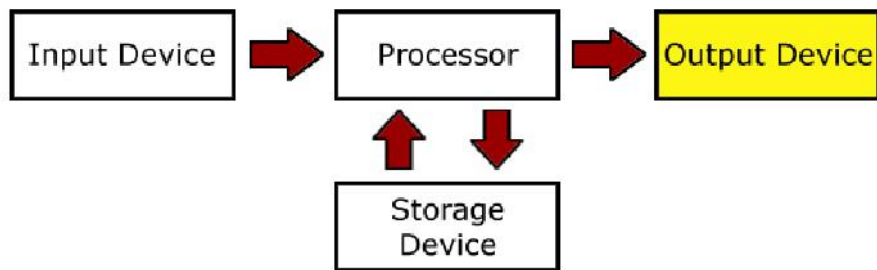
Overview of Output Devices:

Output devices are part of the 4 main hardware components of a computer system:

- **Input Devices** - these get data into the computer
- **Processor** - this does something with the data to make it useful information
- **Output Devices** - these show the results of processing
- **Storage Devices** - holds the data in the system

The image below shows where output devices fit into a computer system:

Examples:



This diagram shows where output devices fit into a computer system.



A computer monitor (screen) is an example of a commonly used output device.

Some devices produce a **temporary output**.

For example: a computer monitor produces images on the screen that constantly refresh and change.

Some devices produce a **permanent output**.

For example: a printer produces outputs as hard copies on paper (these cannot be changed once printed).



Printers are another example of a commonly used output device.

The information below discusses some examples of output devices including their uses, advantages and disadvantages.

NOTE:

More examples will be shown further down this page.

Examples of Output Devices



Key Words:

CRT Monitor , TFT Monitor, Laser Printer, Inkjet Printer, Dot matrix Printer, Plotter, Speakers, Multimedia Projector

The table below shows you some examples of output devices. Click each device to move to more information:

CRT Monitor 	TFT Monitor 	Laser Printer 	Inkjet Printer 	REMEMBER! Output devices send processed information OUT of a computer. NOTE: Click the images to the left to move to more information about each one.
Dot Matrix Printer 	Plotter 	Speakers 	Multimedia Projector 	

Breakdown of each output device:

The following sections will help summarise each of the manual input devices. For each one you will see:

- **Information about the device**
- **Uses of the device**
- **Advantages of the device**
- **Disadvantages of the device**

1 CRT Monitor (cathode ray tube)

Information:

CRT stands for '**cathode ray tube**'.

These are the **old, bulky, heavy** monitors that are still in use today (although being used less and less).

Examples:

These are the **least expensive** types of monitor.

CRT's are becoming **rare** as TFT monitors are replacing them.

Note: TFT's are replacing CRT's largely because they **produce a higher quality** screen image and take up far **less space**.

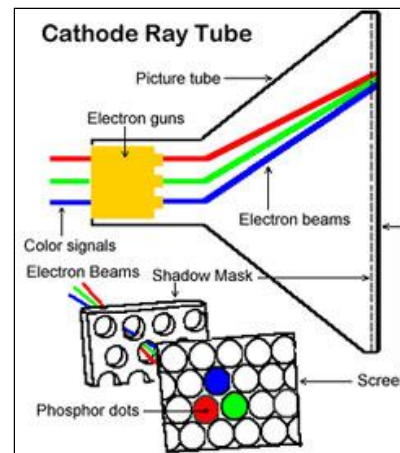
CRT's produce an image (Made up of tiny dots) by firing **Electron Beams** against a **Phosphor Screen**

Each dot is coloured **Red**, **Green** or **Blue** of different intensities which produces a vast amount of different colours.

Note:

These dots **combine** to create millions of different colour shades.

(just like when you mix colours in art)



CRT monitors are very bulky and take up a lot of desk space.



Light pens can be used with CRT's to produce drawings and designs directly onto the screen.

Uses of a CRT Monitors:

CRT Monitors allow users to **instantly see outputs** from the computer.

For example - Users would be able to see words they are typing into word processors.

Users could also see changes that they are making to an image in graphics editors (like Photoshop)

CRT's are also used with **light pens** to allow drawings to be created on screen.



Screen glare can be a problem with CRT's.

Advantages/Disadvantages of CRT Monitors:

Advantages	Disadvantages
Can be used with light pens to create drawings on screen. Note: light pens and CRT's are also used with CAD software (computer aided design).	They are very large and bulky which means they take up large amounts of desk space. Note: their large size and weight also make them difficult (and dangerous) to move.
Produce a larger range of colours than TFT monitors.	They create a lot of heat and have been known to cause fires.
The screen angle is better than with TFT's. (you do not have to stand directly in front of the CRT screen in order to see what is on it)	Screen glare when in direct light makes viewing difficult. Note: they are also prone to ' screen flicker ' which can cause headaches and eye strain.
Cheaper to buy than TFT screens.	They also need more power than TFT's to operate (more expensive to run).



CRT monitors can get very hot.

2 TFT Monitors (thin film transistor)

Information:

- # TFT stands for '**thin film resistor**'.
- # These are **thin flat screened monitors** that are commonly seen today.
- # As they become cheaper, these have taken over from CRT monitors as the most popular

Examples:

computer screens.

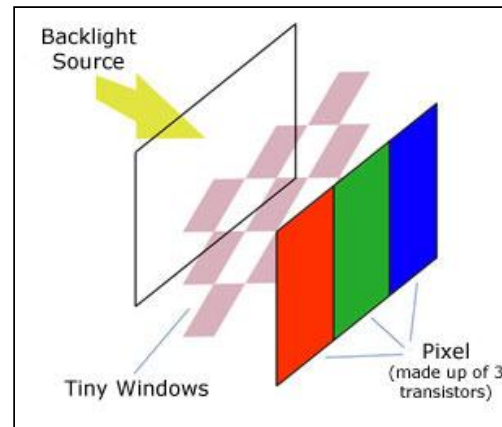
TFT's are used with the **newer desktop PC's, laptops and mobile phones.**

TFT screens are made up of **thousands of tiny pixels**. Each pixel has **3 transistors - red, green and blue** and each transistor can produce different intensities.

This allows huge amounts of colours to be produced (16.7 million).

A **back light** is used to shine light onto these pixels in order for the colours to be produced.

In order to light a single pixel, **tiny windows** can be **opened** and **closed** to allow the light to pass through and onto the pixel.



TFT monitors are much thinner than CRT's.



TFT screens are light and are perfect as displays for portable devices such as laptops.

Uses of TFT Monitors:

TFT Monitors allow users to **instantly see outputs** from the computer.

They are used as **screens** in **laptops** and also **mobile phones**.

Their **thin** and **light design** helps the laptop remain **portable**
(light and easy to transport)

Advantages/Disadvantages of TFT Monitors:

Advantages	Disadvantages
Lightweight so they are perfect for portable laptops and wall mounting.	Images can appear unclear when viewed from an angle .
They are very thin and do not take up as much desk space as a CRT monitor.	(you need to look straight at the screen for the best image)
They produce much less glare than CRT's and so are friendly to the eye.	Lower picture quality (definition) compared to a CRT monitor.
They use less power than CRT monitors (cheaper to run).	Cannot be used with Light Pens yet. (Computer Aided Drawings are not yet possible on TFT monitors)
They emit less radiation than CRT monitors.	
They produce less heat than CRT's.	



TFT's are also used as displays in mobile phones.



Their lightweight makes TFT's easy to mount on walls.

3 Laser Printer

Information:

- # These produce **high quality hard copy** outputs (on paper) .
- # Laser printers use **Toner** pressed on a page to produce printed documents.

Examples:

- # They can print **large amounts** of pages at a **very quick rate**.
- # You can buy both **black and white** and **colour** laser printers.
- # They use a **buffer memory** which stores the information for the whole document before the pages can be printed out.

For example: Once you press print, the document is sent to the printers memory (**RAM**) and stored there. The printer will then output whatever is in the memory when it is ready to do so.

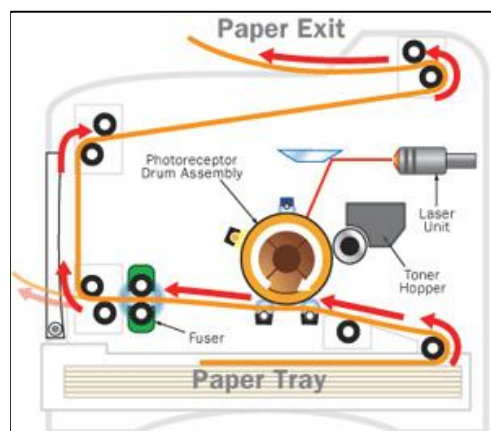
Think of the Buffer Memory as a **kind of waiting room**

- # Laser printers work by using a **laser beam** to 'draw' onto a drum.

The laser leaves a **positive electric charge** on the drum in the places where it drew.

A special power known as '**toner**' is sprayed onto the drum and it sticks to the parts that contain the positive electric charge.

The drum is then **pressed** onto a sheet of paper and the paper is heated (**fused**) so that the toner is bound to the paper.



Laser printers are popular for environments that require fast high-quality printouts.



Laser printers can handle very large print jobs.

Uses of Laser Printers:

- # Very useful for printing where **noise levels** need to be kept **low** (Like in an office where people need to concentrate or use telephones etc).

- # Good at producing **high quality printouts** very **fast**.
- # Good for environments where **high volume of printing** is required (lots of printouts).
This is because they have a **low printing cost per page**.

Their **fast printing speed /low printing cost per page** makes them the **printer of choice on networks**. This is because they need to be able to keep up with the printing demands of many computers.

Popular with:

- **Schools**
- **Businesses**
- **Offices.**



Colour laser printers are expensive to buy and run.



Toner cartridge (these contain toner power).

Advantages/Disadvantages of Laser Printers:

Advantages	Disadvantages
Very fast at printing documents.	Environmentally unfriendly (produce ozone). They also use more power than inkjets.
Toner cartridges last a long time and don't need to be replaced often.	Laser printers are more expensive to buy (but cheap to run).
Laser printers can handle high volumes of printing.	Colour laser printers are much more expensive to buy and run than inkjets.

Very quiet when printing.

Very **high quality** printouts.

Reliable - don't breakdown as easily as inkjet printers.

Cheaper to run than inkjets (low cost per page).

They are **larger in size** compared to other types of printers.

4 Inkjet Printer

Information:

- # These are used to produce **high quality hard copies** (on paper).
- # Although the quality of Inkjet printouts is not as good as Laser printers, it is far better than Dot Matrix printers.
- # Ink is stored in and delivered to the paper via an **ink cartridge**.
- # Inkjet's can print in both **black and white** and **colour**.
- # Inkjet printers have **small amounts of internal memory** (RAM) to store print jobs so printing is done a bit at a time.

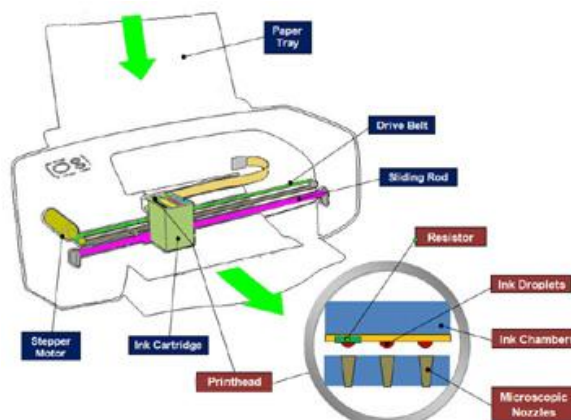
The whole document cannot be stored in printer's RAM so there is often a **pause** while the computer sends the rest of the data

- # Inkjet printers work by **spraying ink** onto a page of paper.

The ink is held in 'ink cartridges' and there are **4 different colours** - **black**, **cyan**, **magenta** and **yellow**.

These mix to form a vast amount of other colours.

Motors are used to position the ink cartridges over the correct position of the page and to drag the paper through the



Examples:



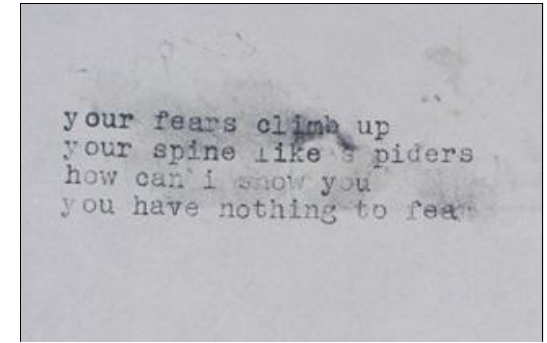
Inkjets can produce photo-quality colour printouts.



Inkjets require 4 separate ink cartridges - Black, Cyan, Magenta, Yellow.

printer.

Click image to zoom



Inkjet printouts will smudge if not left to dry before handling.

Uses of Inkjet Printers:

- # Good where **few printouts** are needed (low volume).
- # Ideal for **single-page**, very high quality printing. (Photo quality images).
- # Often used to print **photographs**.

Advantages/Disadvantages of Inkjet Printers:

Advantages	Disadvantages
Output is very high quality (Especially good for colour photograph printing).	Slow printing rates , especially if several copies are needed.
Much cheaper to buy than laser printers.	Ink cartridges don't last long and so are not suitable for large print jobs.
Smaller than laser printers so take up less space.	They can be expensive to run as new ink cartridges are not cheap to buy.
Do not produce ozone or other harmful compounds.	Printouts can easily smudge if document is not left to dry properly.
Very quiet whilst printing.	

5 Dot Matrix Printer

Information:

- # Often called an '**Impact Printer**' where pins are pressed against an inked ribbon which produces an imprint on paper.
- # They are very **slow**, **noisy** and produce **poor quality** outputs (click [here](#) for video).

Examples:

- # Are still useful where '**Continuous Paper**' needs to be used. For example, printing off all of a business's employee payslips.

Continuous paper is the name given to hundreds of **sheets of paper** that are **joined together** and '**continuously**' fed through the printer.

The paper sheets can be separated after printing is complete.



Dot matrix printers are noisy and slow.

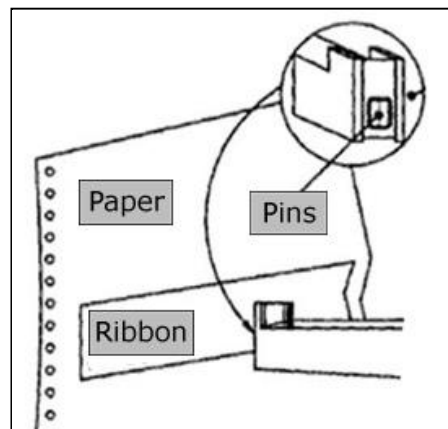
- # Dot matrix printers are still used in environments **where print speed and noise is not an issue** (like in car garages and factories).

- # These printers work by using a **set of pins** to press an **inked ribbon** against **paper**.

When the pin/ribbon hits the paper a **printed dot** is left behind.

The **dots can be arranged** to form **basic text** and **images**.

Different coloured ink ribbons can be used to produce colour printouts.



Dot matrix printers use ink ribbons.

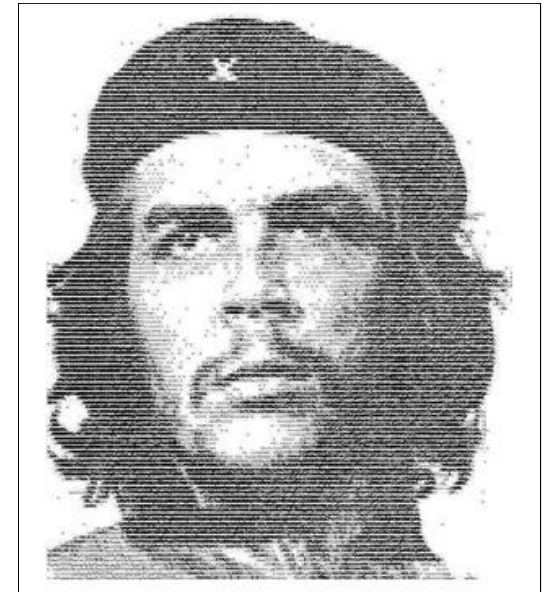
Uses of Dot Matrix Printers:

- # Dot matrix printers are **very robust** and can be used in **noisy and dirty environments** (like garages and warehouses).

- # Can make '**Carbon Copies**' of a document using special '**Carbon Paper**' underneath normal printer paper.

Advantages/Disadvantages of Dot Matrix Printers:

Advantages	Disadvantages
Work just fine in dusty and dirty environments where laser or inkjet printers could easily be damaged.	Very noisy so not suitable for an office environment
Carbon copies can be produced using carbonated paper.	Very slow to print (slowest of all the types of printers)
Very cheap to buy and maintain. (Inked ribbons are very cheap)	Very poor print quality.
Good for continuous printing . (e.g. long print jobs like wage slips)	



Dot matrix printout quality is very low.



Dot matrix printouts are made up of arranged dots.

6 Plotter

Information:

- # There are 2 different types of plotters --- **drum** and **flatbed**.
- # Produce hard copies but are capable of **printing on very large** pieces of paper (e.g. 3 foot wide by 10 foot long).

Examples:

- # The print quality of plotters is **very high**.
- # Plotters can also produce **very large and accurate documents** such as blueprints (plans) for buildings
- # Plotters **draw images** on the page using **coloured pens** which are moved around the paper by a computer.



A drum plotter.



A flatbed plotter.



Plotters can produce very large printouts like those seen on advertising boards.

Uses of Plotters:

- # Used to produce **large, accurate drawings**. Some examples are listed below:
 - **Blueprints of buildings** (plans/designs)
 - **Maps**
 - **Giant posters and adverts** (like those we see on billboards).
- # Pens can be replaced with **cutting tools** to produce **large signs**.

Advantages/Disadvantages of Plotters:

Advantages	Disadvantages
They can produce much larger printouts than other printers.	Their print times are very slow .
The print quality is extremely high .	They are very expensive to buy and maintain.
Plotters can also 'cut out' designs for use in sign making.	

7 Speakers

Information:

Many everyday uses of a computer require the user to listen to different types of sounds. For example:

- **Listening to a music album**
- **Watching a movie.**
- **Browsing videos on YouTube**
- **Multimedia encyclopedias.**

In order for the computer to output these sounds it requires some sort of **speaker system**!

Speakers can be connected directly to a computer or are built into the monitor or casing.

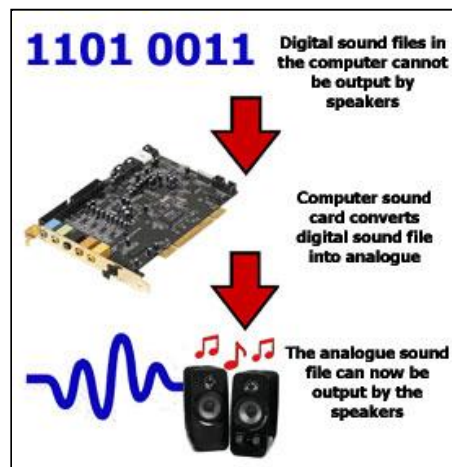
Note: Most computers have small **built in speakers** to alert users when they do something wrong but these can only produce very **basic sounds** like 'beeps'.

Speakers are **analogue** and cannot output digital 'computerised' sounds.

Digital data from the computer must first be **converted into analogue** sound using a **sound card**.

The analogue sound can then be output via the speakers and we can listen to it.

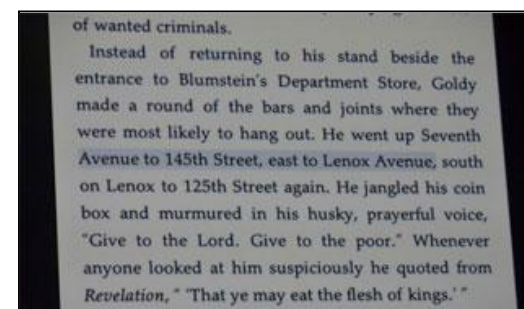
Speakers can also be used to **alter the volume** of the sounds.



Examples:



Speakers allow us to listen to sound coming out of a computer system.



E-Book readers (like the Amazon Kindle) have special software called 'screen readers' that can narrate the words on the screen.



Speakers are an important part of home entertainment

Uses of Speakers:

- # Used to output sound from **multimedia presentations** (like famous speeches etc).
- # Used in **home entertainment** (surround sound etc).

systems.

- # Can be used by **blind people** to **listen to words** on the screen instead of reading .

Note: Special software called '**screen readers**' are also needed in order to convert the printed words into sound.

- # Can be used to play downloaded **music** files.

Advantages/Disadvantages of Speakers:

Advantages	Disadvantages
Everyone in the room can hear the sound.	Output from speakers can disturb people trying to work.
They can help blind people use a computer because text can be turned into sound.	High quality speakers can be expensive .

8 Multimedia Projectors

Information:

- # These **project** whatever appears on a computer monitor onto a **very large screen** so that large numbers of people can view it.
- # Multimedia Projectors are usually controlled using a **remote control**.
- # The remote control makes it possible to direct the presentations without the need to be at the computer.
- # Multimedia projectors are often **hung from the roof** on special brackets.

Examples:



A multimedia projector.

Uses of Multimedia Projectors:

- # Can be used for **training presentations** to allow the whole audience to see images from a single computer.

Also used for **large scale advertising** of new products.

For example: the famous Apple Technology Innovations presentations with Steve Jobs (click [here](#) to watch)



Multimedia projectors can be used to project images onto a very large screen.

Home cinema systems where images from DVD's or televisions are projected to a large screen.

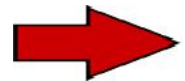
Advantages/Disadvantages of Multimedia Projectors:

Advantages	Disadvantages
Many people to clearly view the same presentation.	Lower image quality compared to a computer screen.
Enhance the viewers experience.	Very expensive to buy.
For example: watching movies on a large screen is better than a small TV screen.	Cooling fans in the projector are noisy .



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